

Surrogate Modeling for Scalable Fast Time-to-Solution Model Exploration of Stochastic Simulations

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■ Context

- *Broader*: Surrogate modeling for stochastic computer models
- *Focused*: Agent-based models, fast time to solution

■ Approach

- Gaussian process-based surrogates
 - Quantile GP
 - Cluster GP
 - Trajectory-oriented Common Random Number GP
- Fast time-to-solution
 - Choice of surrogate
 - Choice of acquisition function (**Binois et. al** [1])
 - Robust, scalable computing infrastructure

Broader Context

■ Computer simulation

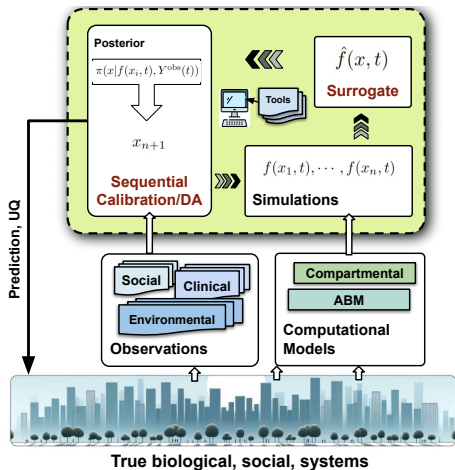
Encode the underlying dynamics (**epidemic process**) to study and predict its behavior.

■ Stochastic model

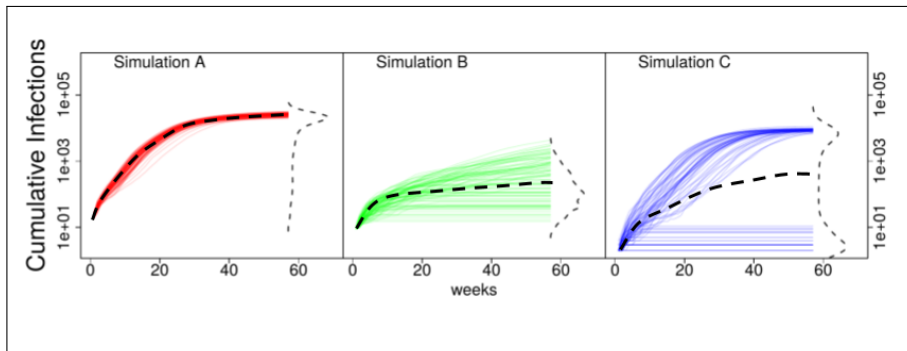
Encode uncertainties in underlying processes through randomness (e.g., **seeding infections, specific mixing patterns**).

■ Statistical Inference

Tools to calibrate the model to empirical reality, extra attention needed for stochastic model.

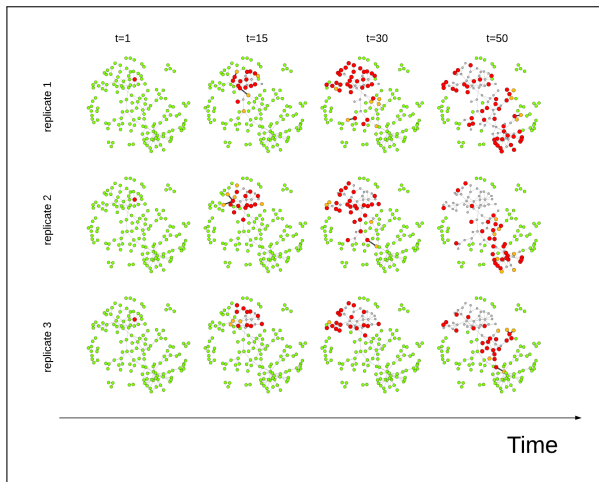


Focused Context – Surrogates



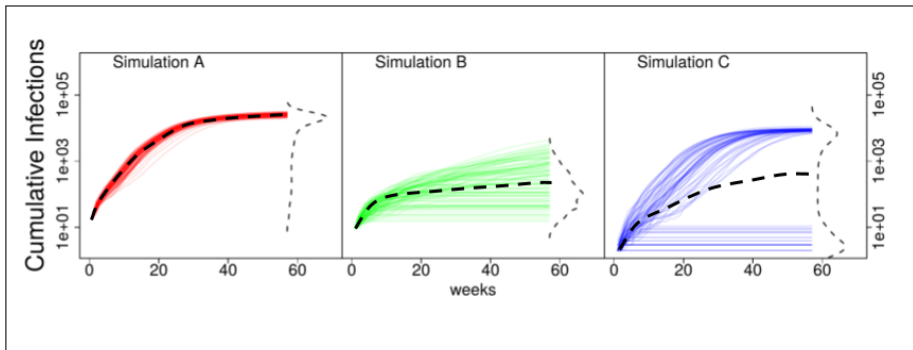
- Fast turn-around time
 - Expensive-to-run, agent-based simulation → **Surrogate assisted Strategic sampling**
- Ability to capture detailed model states for online (i.e., sequential in time) analysis

Focused Context – challenges in surrogate modeling



- Each simulated scenario defines a unique **population mixing** in the heterogeneous contact network, even at the same disease-specific parameter values.
 - Possibly infection seeding too.

Focused Context – challenges in surrogate modeling



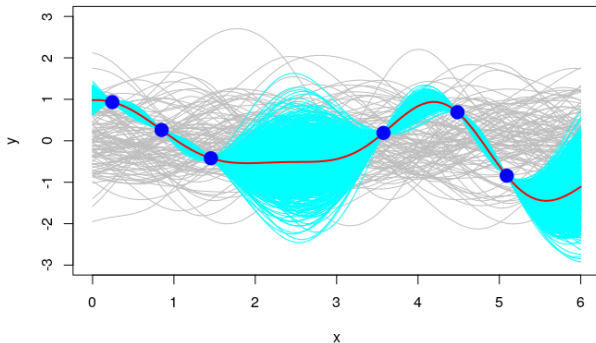
- The replicate variability is not Gaussian.
- Also not homoskedastic.
- Interested in multivariate/vector response.

Statistical problem

- Fit a surrogate model $\eta(\theta)$ (using a **Gaussian process**)
 - Quantile GP [4] –
summarize the replicate variability via **quantiles**
 - Cluster GP –
use independent GPs for **clusters** of simulator response
 - Common random number GP [3] –
emulate individual replicates by exploiting **common random number**

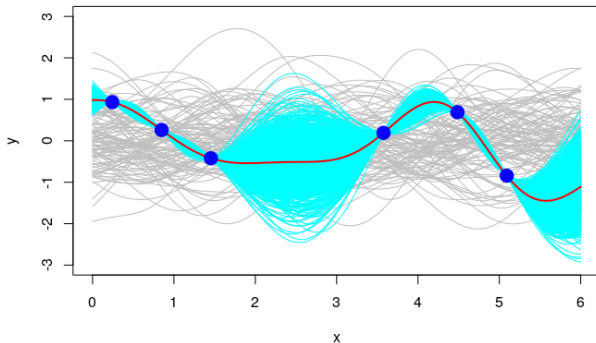
Brief introduction to Gaussian Process as surrogate

- **Input:** $\theta = (\theta_1, \dots, \theta_N)^T \Leftrightarrow$ (transmissibility, # initial infections, ...)
- **Response:** $\eta = (\eta_1, \dots, \eta_N)^T \Leftrightarrow$ (deaths, hospitalizations, ...)
- **GP prior:** $\eta(\cdot) \sim GP(\mu(\cdot), k(\cdot, \cdot; \phi)) \Rightarrow \boxed{\eta \sim MVN(\mu, K_N + \Sigma_N)}$
where, $K_{ij} = k(\theta_i, \theta_j; \phi)$



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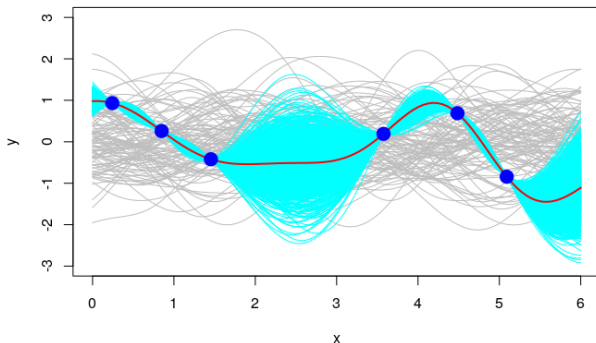
Brief introduction to Gaussian Process as surrogate

■ **Prediction:** $\eta(\theta^*) | \boldsymbol{\eta} \sim N(\mu(\theta^*), \sigma^2(\theta^*))$

$$\mu(\theta^*) = \mathbf{k}(\theta^*)^\top (\mathbf{K}_N + \Sigma_N)^{-1} \boldsymbol{\eta},$$

$$\sigma^2(\theta^*) = k(\theta, \theta) + \tau^2 - \mathbf{k}(\theta^*)^\top (\mathbf{K}_N + \Sigma_N)^{-1} \mathbf{k}(\theta^*),$$

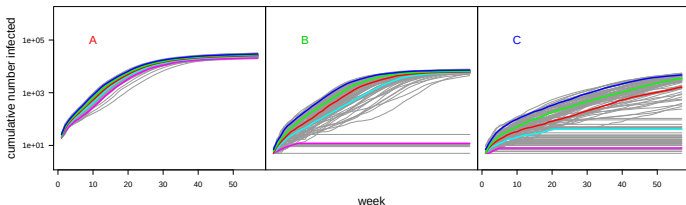
where, $\mathbf{k}(\theta^*) = (k(\theta^*, \theta_1), \dots, k(\theta^*, \theta_N))^\top$



Quantile GP

Preprocess the data - empirical quantiles

- Pointwise $\alpha = (.05, .275, .5, .725, .95)$ -quantiles estimated from the 100 replicates.
- Use α as an additional input to GP.
- α -quantile estimate may not correspond to an actual simulation trajectory.

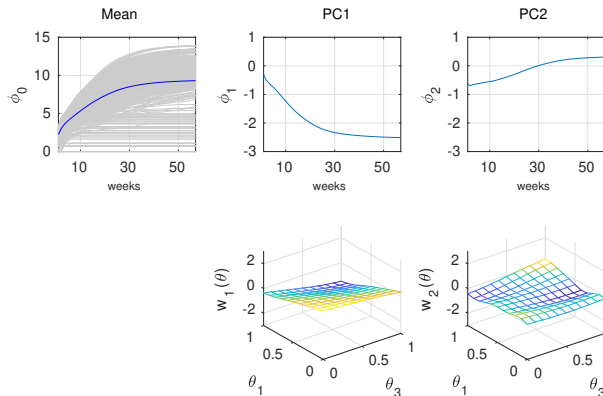


Simulated trajectories in grey, and estimated quantiles in color.

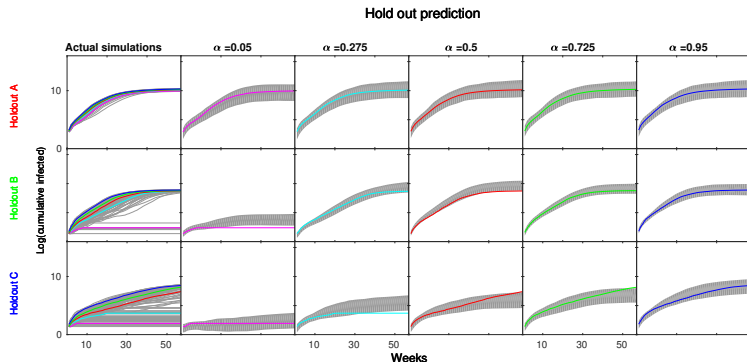
Quantile GP on multivariate output

Basis representation of multivariate output using principal component

$$\eta(\theta, \alpha) = \phi_0 + \sum_{k=1}^{p_\eta} \phi_k w_k(\theta, \alpha) + \epsilon_{w0},$$



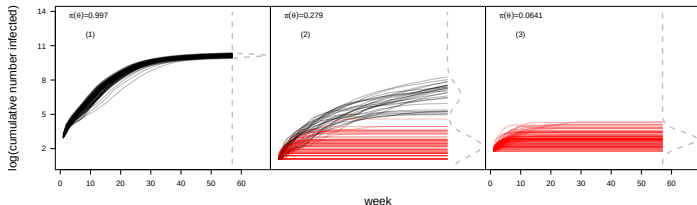
Quantile GP



The first column from the left shows the actual simulations and their empirical quantiles at the three design inputs denoted by their colors from the design. Next 5 columns show the pointwise 90% credible intervals of the posterior emulator prediction at those model inputs.

Cluster GP

Instead, we may consider a mixture of Gaussian.

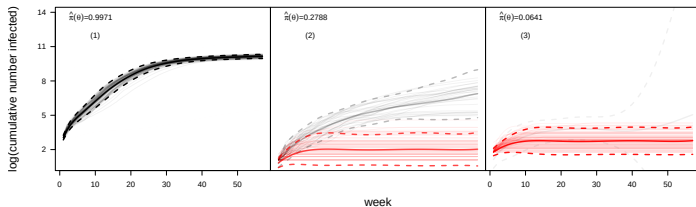


i.e.

- Consider that the simulation output corresponding to an input falls into either of two categories - (a) epidemic, or (2) no epidemic.
- Gaussian assumption may make sense now in each category.

Cluster GP

... and, use two separate GPs to model the simulation output.



Full model:

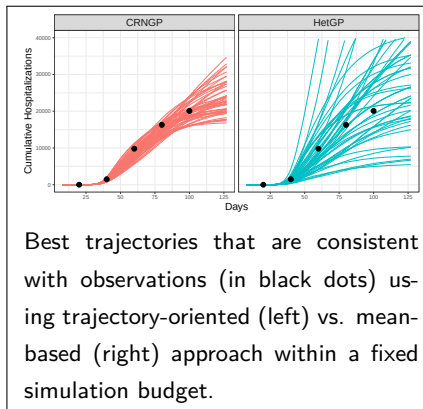
$$\eta(\boldsymbol{\theta}, t) = I_{[\pi(\boldsymbol{\theta}) > 0.5]} w_1(\boldsymbol{\theta}, t) + I_{[\pi(\boldsymbol{\theta}) < 0.5]} w_2(\boldsymbol{\theta}, t)$$

$$\pi(\boldsymbol{\theta}) = \frac{\exp(z(\boldsymbol{\theta}))}{1 + \exp(z(\boldsymbol{\theta}))}$$

$$z(\boldsymbol{\theta}) \sim GP(0, \Sigma(\boldsymbol{\theta}, \boldsymbol{\theta}'))$$

$$w_k(\boldsymbol{\theta}, t) \sim GP(\mu_k, \Sigma_k((\boldsymbol{\theta}, t), (\boldsymbol{\theta}', t'))), \quad k = 1, 2$$

Challenges in using Quantile or Cluster GP as surrogate



- Each simulated trajectory defines a unique **population mixing** in the heterogeneous contact network, even at the same disease-specific parameter values.
 - Possibly infection seeding too
- There is value in knowing individual trajectories instead of just the **good parameter values** for the simulation in an average sense.

Common Random Number GP (CRN-GP)

Basic idea of CRNGP

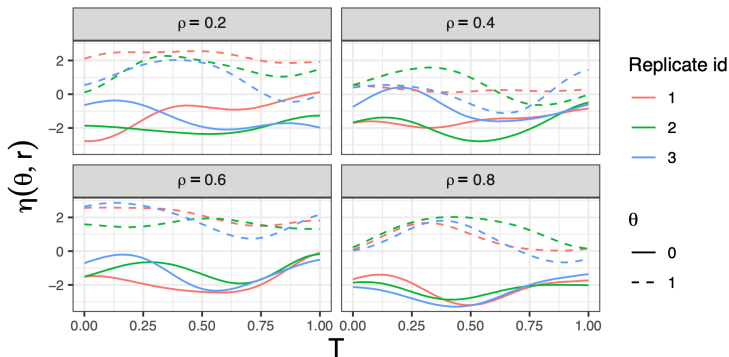
- Define a GP prior, that can explore **similarities** in random seed:

$$\eta(\boldsymbol{\theta}, r) \sim GP(0, \mathbf{k}(\boldsymbol{\theta}, r))$$

- Because $\eta(\boldsymbol{\theta}, r)$ represents time-series outputs, we further index by t and model $\eta(\boldsymbol{\theta}, r, t)$.
- Define **CRN covariance kernel**. Given a $0 < \rho < 1$,

$$\mathbf{k}((\boldsymbol{\theta}_i, t), r), ((\boldsymbol{\theta}_j, t'), r')) = \begin{cases} \mathbf{k}((\boldsymbol{\theta}_i, t), (\boldsymbol{\theta}_j, t')) & \text{if } r = r' \\ \mathbf{k}((\boldsymbol{\theta}_i, t), (\boldsymbol{\theta}_j, t')) \times \rho & \text{o.w.} \end{cases}$$

CRN-GP



Realizations from CRNGP for different values of ρ (keeping the other GP hyperparameters fixed). Each line corresponds to a random realization from the CRNGP which is vector output (trajectory) indexed by variable T (scaled to $[0, 1]$).

CRN-GP

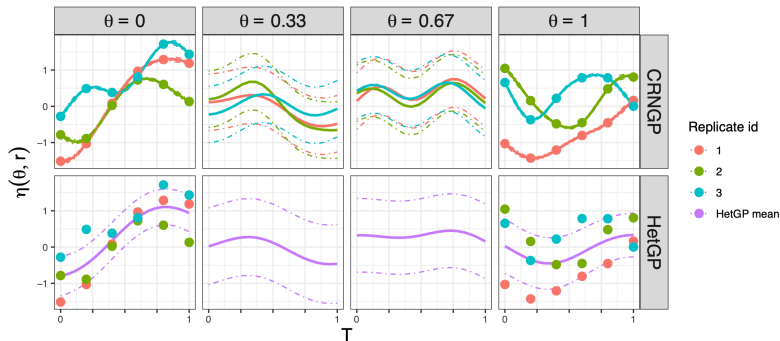


Figure: Comparison between fitted CRNGP (top) and Heteroskedastic GP (bottom) on vector-valued response at 4 different values of input x . Each color corresponds to a random seed value except the purple which represents the predicted mean surface from the HetGP model.

Thursday, February 29

MS152

New Approaches to Uncertainty Quantification in Surrogate Modeling and Inverse Problems

2:30 PM - 4:30 PM

Room: Riviera - Hotel Savoia Excelsior Palace

This minisymposium aims to showcase the most recent advancements in mathematical and statistical methodologies, along with innovative applications, focused on uncertainty quantification in surrogate modeling and inverse problems. The speakers in this minisymposium encompass a range of experience levels, including junior, mid-career, and senior researchers in the fields of mathematics and statistics. They will share their latest research across a diverse array of topics, including dimension reduction, machine learning techniques, simulation-based experiments, and real data analyses in remote sensing. This will address both the challenges and new discoveries in the realm of uncertainty quantification.

Organizer: Emily L. Kang

University of Cincinnati, U.S.

2:30-2:55 Reinforcement Learning for the Adaptive Construction of Surrogate Models [abstract](#)

Nick Winovich, Sandia National Laboratories, U.S.

3:00-3:25 Trajectory-Oriented Optimization of Stochastic Epidemiological Models [abstract](#)

Mickael Binois, Inria, France; Arindam Fadikar, Nicholson Collier, and Abby Stevens, Argonne National Laboratory, U.S.; Kok-Ben Toh, Northwestern University, U.S.; Jonathan Ozik, Argonne National Laboratory, U.S.

3:30-3:55 Learning-Based Estimation and Uncertainty Quantification of Nonlinear Inverse Operators [abstract](#)

Rick Lucas and Emily L. Kang, University of Cincinnati, U.S.; Jonathan Hobbs, NASA, Jet Propulsion Laboratory, U.S.

Requirements for An Open Science Platform for Robust Epidemic Analysis

Developing Distributed High-performance Computing Capabilities of an Open Science Platform for Robust Epidemic Analysis

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IEEE Workshop on Parallel and Distributed Processing for Computational Social Systems (ParSocial 2023), IPDPS Workshop, May 15-May 19 2023, St. Petersburg, Florida USA [2]

OSPREDY Goals and Requirements

■ Integrated, algorithm-driven HPC workflows (Goal 1)

- Fast time-to-solution workflows
 - Analyses need to provide actionable insights quickly
- Coordinated multi-resource task execution
 - Epi-modeling needs a range of tasks to be executed across distributed/heterogeneous ecosystem resources
- Robust, secure, and automated access to federated HPC resources
 - Wide range of access, authentication, security protocols
- Scalable, fault-tolerant task execution
 - Demand can vary dramatically over time
- Multi-language workflows
 - Need inclusive, multi-language APIs
- Efficient wide-area data staging
 - Uniform access to epi-data and model artifacts

OSPREDY Goals and Requirements (cont.)

■ Data ingestion, curation, and management (Goal 2)

- Data stream ingestion
 - Move and track data sets from origin to their site of use
- Automated data curation
 - Data analysis pipelines for data de-biasing, data integration, uncertainty quantification, and metadata and provenance tracking
- Managing algorithm and model artifacts
 - Serve algorithm and model artifacts for, e.g., data assimilation

■ Shared Development Environment for rapid response and collaboration (Goal 3)

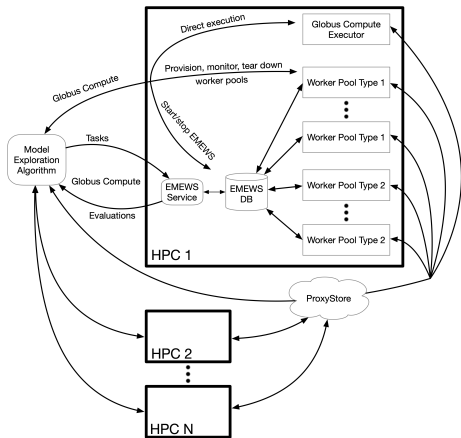
- Model and workflow sharing
 - Portable workflows that run on federated HPC systems users have access
- Model validation and publishing
 - Post complete models with the data used to validate them for reproduction, extension, or scaling by others

Prototype OSPREY Architecture

- The OSPREY architecture includes

- Model exploration module
- Function as a Service (FaaS) task distribution
- EMEWS task DB
- Scalable worker pools
- Efficient data distribution

- These collectively provide the ability to **robustly and securely** coordinate and scale (epidemic) workflows across a distributed ecosystem of heterogeneous remote resources



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- NIH R01DA055502: Leveraging data synthesis to identify optimal and robust strategies for HIV elimination among substance-using MSM
- Argonne Leadership Computing Facility
- Laboratory Computing Resource Center at Argonne National Laboratory